



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2022 Aug: IIT M FOUNDATION DIPLOMA 2022 Aug: IIT M

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

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TIMER

00:19



CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653357062

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "FOUNDATION LEVEL: SEMESTER 2/DIRECT ENTRY DIPLOMA : MATHEMATICS FOR DATA SCIENCE 2"



ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ Yes☐ No

Your score : 0

Discussions (0)



Question 2 : 640653357063

Total Mark : 0.00 | Type : COMPREHENSION

Consider a matrix $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $f(a, b) = \det(A)$. Answer the given subquestions:



Your score : 0



Question 3 :
640653357064



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the number of critical points of the function $f(a, b)$.

Answer (Numeric):

Accepted Answer : 1

Your score : 0

Discussions (0)



Question 4 :

640653357065

View Parent QN

View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Which of the following options is/are true?

OPTIONS :

☐ A is not a symmetric matrix for all $a, b \in \mathbb{R}$.

☐ A^2 is a symmetric matrix for all $a, b \in \mathbb{R}$.

☐ If (β, γ) is a critical point of $f(a, b)$, then the matrix $A = \begin{bmatrix} \beta & \gamma \\ \gamma & \beta \end{bmatrix}$ satisfies that, $A^2 = A$.

☐ If (β, γ) is a critical point of $f(a, b)$, then the matrix $A = \begin{bmatrix} \beta & \gamma \\ \gamma & \beta \end{bmatrix}$, $\text{rank}(A) = 1$.

Your score : 0

Discussions (0)



Question 5 : 640653357066

Total Mark : 0.00 | Type : COMPREHENSION

Consider the matrix $A = \begin{bmatrix} x & -1 & 2 \\ y & 1 & 2 \\ z & 3 & 4 \end{bmatrix}$. Answer the given subquestions:

Your score : 0



Question 6 :
640653357067

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a linear transformation such that $T(x, y, z) = \det(A)$. Then which of following options is/are true?



OPTIONS :

- ☐ T is injective.
- ☐ T is surjective.
- ☐ T is an isomorphism.
- ☐ T is neither injective not surjective.

Your score : 0

[Discussions \(0\)](#)

Question 7 :
640653357068

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Let W be a subspace of $\mathbb{R}^3$ (with inner product as the dot product) such that $W = \{(x, y, z) \mid \det(A) = 0\}$. If β is an orthonormal basis of W , then find the cardinality of the set β .



Answer (Numeric):

Accepted Answer : 2

Your score : 0

 Discussions (0)

Question 8 :

640653357069



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be a linear transformation such that $T(x, y, z) = \det(A)$. Let B be the matrix representation of T with respect to some ordered bases for the domain and the codomain. Let $m \times n$ be the order of the matrix B and r be the nullity of B . Then find the value of $m + n + r$.



Answer (Numeric):

Accepted Answer : 6

Your score : 0

 Discussions (0)

Question 9 : 640653357070

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation defined by
 $T(x, y, z) = (x + 2y, z - 3y, x - y + z)$.



Your score : 0



Question 10 :
640653357071



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the nullity of T .



Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

Discussions (0)



Question 11 :
640653357072



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the rank of T .



Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)

**Question 12 :**
640653357073

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Choose the correct option(s).

OPTIONS :

☐ $\{(-4, 2, 6)\}$ is a basis for the kernel of T . ☐ $\{(0, 1, 3), (-2, 1, 0)\}$ is a basis for the kernel of T . ☐ There exists an isomorphism from the range of T to $\mathbb{R}^2$. ☐ There exists an isomorphism from the range of T to $\mathbb{R}$.

Your score : 0

Discussions (0)

**Question 13 : 640653357074**

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

An inner product on a vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying the following conditions: Condition 1: $\langle v, v \rangle > 0$ for all $v \in V \setminus \{0\}$; $\langle v, v \rangle = 0$ if and only if $v = 0$.Condition 2: $\langle v_1 + v_2, v_3 \rangle = \langle v_1, v_3 \rangle + \langle v_2, v_3 \rangle$.Condition 3: $\langle v_1, v_2 \rangle = \langle v_2, v_1 \rangle$.Condition 4: $\langle cv_1, v_2 \rangle = c\langle v_1, v_2 \rangle$ Let $V = \mathbb{R}^2$ and consider the function defined as:

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

$$\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - 4x_2 y_2.$$

Which of the following are satisfied by the above function?

OPTIONS :

- ☐ Condition 1 is satisfied.
- ☐ Condition 2 is satisfied.
- ☐ Condition 3 is satisfied.
- ☐ Condition 4 is satisfied.

Your score : 0

Discussions (0)



Question 14 : 640653357081

View Solutions (0)

Total Mark : 2.00 | Type : MSQ

Which of the following options is/are true?

$$f(x, y) = \frac{xy^3}{x^2 + y^6}$$

OPTIONS :

- ☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$
- ☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \frac{1}{2}$
- ☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$
- ☐ $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.
- ☐ $f(x, y)$ is not continuous at $(0, 0)$.
- ☐ $f(x, y)$ is continuous at $(0, 0)$.

Your score : 0

Discussions (0)



Question 15 : 640653357093

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ -1 & 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 3 & 4 \\ 0 & 2 & 1 \\ -1 & 1 & 3 \end{pmatrix}$. Choose the correct option(s).

OPTIONS :

☐ A is not equivalent to B. ☐ A is equivalent to B. ☐ A is similar to B. ☐ A is not similar to B. 

Your score : 0

[Discussions \(0\)](#)

Question 16 : 640653357075

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

If a , b and c are three positive numbers which satisfy the following two properties:

- The sum of a , b and c is 27.
- The sum of the squares of a , b , c is minimum among the sum of squares of any such positive numbers which sum up to 27.

Find the value of $a - b + c$.

Answer (Numeric):

Accepted Answer : 9

Your score : 0

[Discussions \(0\)](#)**Question 17 : 640653357076**

Total Mark : 0.00 | Type : COMPREHENSION

Consider the function $f(x, y) = 2x^3 + 6xy^2 - 3y^3 - 150x$.
Answer the given subquestions:



Your score : 0

**Question 18 :
640653357077**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find the number of local maxima using the Hessian test.

Answer (Numeric):

Accepted Answer : 1

Your score : 0

[Discussions \(0\)](#)**Question 19 :
640653357078**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find the number of local minima using the Hessian test.

Answer (Numeric):

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 20 :
640653357079



View Parent QN



View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the number of saddle points using the Hessian test.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

 Discussions (0)



Question 21 :
640653357080



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the number of points at which the Hessian test is indeterminate.

Answer (Numeric):

Answer

Accepted Answer : 0

Your score : 0

 Discussions (0)



Question 22 : 640653357094

Total Mark : 0.00 | Type : COMPREHENSION

The price of a product ($f(x, y)$) depends on the price (x) of the raw materials and the price (y) of transportation of the product to the market according to

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$
$$f(x, y) = \begin{cases} x^3 - xy^2 & \text{if } x \neq y \\ x + y & \text{otherwise} \end{cases}$$

Answer the given subquestions using the information above.

Your score : 0



Question 23 :
640653357095

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

What will be the ratio of the price of the raw materials and the price of the transportation ($x : y$) when $y > x$, if the rate of change of the price of the product with respect to the price of the raw materials is 0?
(In this context x and y both are always positive).

OPTIONS :

☐ 1 : 1

☐ 1 : 3

☐ 1 : $\sqrt{3}$

- ☐ The ratio cannot be determined using the given information.

Your score : 0

Discussions (0)



Question 24 :
640653357096

View Parent QN

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Which of the following statements are true?

OPTIONS :

- ☐ $f(x, y)$ is a linear function in its domain.

- ☐ $f(cx, cy) = c^3 f(x, y)$ for any real number c , if $x \neq y$.

- ☐ $f(x, y)$ is continuous at $(0, 0)$.

- ☐ If the price of raw material and transportation of the product approaches 3 and 2 respectively, then the price of the product approaches 15.

- ☐ If the price of raw material and transportation of the product approaches 3 and 2 respectively, then the price of the product approaches 5.

Your score : 0

Discussions (0)



Question 25 :
640653357097

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : SA

If the rate of change of the price of the product along the direction of the vector $(1, 1)$ is $\frac{1}{\sqrt{2}}[ka^2 + lab + mb^2]$, when the price of raw material is a and the price of transportation of the product to the market is b (where $a \neq b$), then find the value of $k - l + m$.



Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

Discussions (0)



Question 26 : 640653357082

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Which of the following is true for the function:



$$f(x, y) = \frac{x^3 + y^3}{x + y}$$

OPTIONS :

☐ $\lim_{(x,y) \rightarrow (1,-1)} f(x, y) = 0$ 

- ☐ $\lim_{(x,y) \rightarrow (1,-1)} f(x,y) = 3$ 
- ☐ $\lim_{(x,y) \rightarrow (1,-1)} f(x,y) = 2$ 
- ☐ $\lim_{(x,y) \rightarrow (1,-1)} f(x,y)$ does not exist. 

Your score : 0

 Discussions (0)



Question 27 : 640653357083

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Consider the function $f(x,y) = x^2 + y^2$. Which of the following affine subspaces represent the tangent line at the point $(1,2)$ in the direction of the vector $(1,1)$? 

OPTIONS :

- ☐ $\{(x,y,z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}}\}$ 
- ☐ $\{(x,y,z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}} = \frac{z-5}{\frac{1}{\sqrt{2}}}\}$ 
- ☐ $\{(x,y,z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-2}{\frac{1}{\sqrt{2}}} = \frac{z-5}{\frac{6}{\sqrt{2}}}\}$ 
- ☐ $\{(x,y,z) \in \mathbb{R}^3 \mid \frac{x}{\frac{1}{\sqrt{2}}} = \frac{y}{\frac{1}{\sqrt{2}}} = \frac{z}{\frac{6}{\sqrt{2}}}\}$ 

Your score : 0

 Discussions (0)



Question 28 : 640653357089

[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MCQ

Which of the following option is/are true?

OPTIONS :

- ☐ If L and L' are two affine subspaces of $\mathbb{R}^3$, then $L \cap L'$ is an affine subspace of $\mathbb{R}^3$.
- ☐ Let $Ax = b$ be a system of linear equations with infinitely many solutions and let P be an invertible matrix such that product PA is well defined. Then the system of linear equations $PAx = b$ has infinitely many solutions.
- ☐ Let A and B be two matrices such that reduced echelon form of A and B are the same. Then $\text{rank}(A) > \text{rank}(B)$.
- ☐ Let $Ax = 0$ be a system of linear equations such that reduced echelon form of the coefficient matrix A is the identity matrix. Then the system $Ax = 0$ has infinitely many solutions.

Your score : 0

[Discussions \(0\)](#)

Question 29 : 640653357084

Total Mark : 0.00 | Type : COMPREHENSION

Consider $V = \mathbb{R}^3$ with the inner product as the dot product and $W = \{(x, y, z) \mid x = 0, y = z\}$ is a subspace of V . Let $P_W: V \rightarrow W$ be a projection on W . Answer the given subquestions:

Your score : 0



Question 30 :
640653357085 View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the dimension of the image space PW.

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

 Discussions (0)**Question 31 :**
640653357086 View Parent QN View Solutions (0)

Total Mark : 2.00 | Type : SA

Find the dimension of the null space of PW.

Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

 Discussions (0)**Question 32 :**
640653357087 View Parent QN View Solutions (0)

Total Mark : 1.00 | Type : SA

If $v \in W$ is such that $\|v\| = 2$, then
find $\|P_W(v)\|$.



Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 33 :
640653357088



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Let A be the matrix representation of P_W with respect to some orthonormal bases β and γ for V and W , respectively. Then find the dimension of the null space A^2 .



Answer (Numeric):

Answer

Accepted Answer : 2

Your score : 0

Discussions (0)



Question 34 : 640653357090

Total Mark : 0.00 | Type : COMPREHENSION

Let $A = \begin{pmatrix} 1 & 0 & 1 \\ -2 & 1 & 2 \\ 1 & 4 & -1 \end{pmatrix}$. Let B be the matrix whose rows are obtained by normalizing the rows of A . Answer the given subquestions about matrix B .



Your score : 0



Question 35 :
640653357091

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Choose the correct options about the matrix B.

OPTIONS :

- ☐ A and B have the same reduced row echelon form.
- ☐ $BB^T x = 0$ has infinitely many solutions.
- ☐ B is an orthogonal matrix.
- ☐ The columns of B are not orthonormal.

Your score : 0

[Discussions \(0\)](#)

Question 36 :
640653357092

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

What is $z_2 + 2z_3$, where $z = (z_1, z_2, z_3)^T$
is the solution of $Bz = (1, -2, 1)^T$
obtained using the Cramer's rule?

Answer (Numeric):

Accepted Answer : 0

Your score : 0

 Discussions (0)



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